

Curves of constant width and a lower bound for Lebesgue's universal covering problem

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Abstract

A universal cover is a convex set in the plane that contains a congruent copy of every planar set of diameter one. Lebesgue asked in 1914 for one of least area, and the value is not known. We prove that every convex universal cover has area at least 0.8344, improving on 0.832, published in 2005, and 0.833, in a 2026 preprint, both of which come from a disc together with an equilateral triangle and a regular pentagon. Our test sets are instead curves of constant width: the disc, the Reuleaux triangle and the Reuleaux pentagon. Each contains the regular polygon it is built on, so the family is strictly larger at the same number of bodies and the same number of placement parameters, and we show that the classical configuration admits an arrangement whose hull has area below 0.8336, so no bound drawn from those three sets by this argument reaches ours. Curves of constant width were proposed for this role, and explored numerically, by Gibbs in 2014; what is added here is a proof. It consists of an analytic reduction followed by one finite computation. The reduction bounds the hull area from below over an entire box of placements at once, by eroding each Reuleaux polygon to a fixed set contained in every placement that box allows. The computation is an exhaustive subdivision of the resulting five-dimensional space, recorded as a certificate of 486,799,600 nodes and checked by a verifier independent of the search, with a rigorous bound on its floating point error six orders of magnitude below the margin the verification attains.

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1 Introduction

A planar set has diameter one when no two of its points lie farther than one apart. Many sets have this property and they look very unlike one another. The disc of diameter one is one of them. So is the equilateral triangle of side one. So is the Reuleaux triangle, obtained from that triangle by replacing each side with a circular arc centred at the opposite vertex. A convex set in the plane is a *universal cover* if every set of diameter one can be moved to fit inside it, by a translation, a rotation, and a reflection where a reflection helps. Lebesgue asked, in a 1914 letter to Pál, how small the area of a universal cover can be. The answer is not known, and the problem is recorded as open by Brass, Moser and Pach [4]. Convexity is part of the question rather than a convenience: Duff [6] constructs a non-convex universal cover of smaller area than any convex one then known, so the convex and unrestricted variants have different answers.

The difficulty is that a single set has to serve all of them at once. Covering any one set of diameter one is easy. The disc of diameter one contains the equilateral triangle of side

one, for instance, but it does not contain the Reuleaux triangle, whose corners reach distance $1/\sqrt{3}$ from its centre while the disc reaches only $1/2$. A cover must find room for both, and for every other set of diameter one, each in some position of its own. Write Λ for the least area of a universal cover, which exists by the Blaschke selection theorem.

Progress on Λ has come from narrowing the interval that contains it, and the two sides of that interval have quite different characters.

Upper bounds are constructive. One exhibits a cover and then removes as much of it as can be spared. Pál [18] starts from the regular hexagon of inradius $1/2$, which is a universal cover, and cuts away two of its corners. Sprague [20] removes a further piece. Hansen [13, 14] removes two very thin slivers. The current record, 0.8440935944, is due to Gibbs [10], and refines the construction that Baez, Bagdasaryan and Gibbs [3] analysed to high precision. Each step in this line is a smaller and more delicate excision than the one before it.

Lower bounds rest on one observation, elementary enough to state in a sentence. Fix finitely many sets of diameter one. A universal cover contains a congruent copy of each of them, and being convex it also contains the convex hull of the union of those copies. The copies may sit anywhere, so the area of the cover is at least the smallest area that hull can have, minimised over every way of placing them. Choosing the sets, and carrying out that minimisation, gives a bound.

The bounds obtained this way have moved much less than the upper bounds. Pál [18] takes a disc and an equilateral triangle, evaluates the minimum by hand, and obtains $\pi/8 + \sqrt{3}/4$. Elekes [7] takes a disc together with regular 3^k -gons and reaches approximately 0.8271. Brass and Sharifi [5] take a disc, an equilateral triangle and a regular pentagon; the placements of three bodies form a five-parameter family once the redundant degrees of freedom are removed, and they minimise over it by subdivision, obtaining 0.832. That bound has stood since 2005 and is still the best that has been refereed. A 2026 preprint of Xie [22] reports 0.833 from the same three sets.

Every one of these bounds comes from a disc together with regular polygons, and the last two come from the same three sets, twenty-one years apart. That pattern suggests a configuration close to exhausted, and we make the suggestion precise rather than leaving it as an impression. The disc, equilateral triangle and regular pentagon admit a placement whose hull has area at most 0.833597897 (Proposition 3.2), and a lower bound drawn from a family can never exceed the minimum hull area of that family. No computation applied to those three sets, however long it runs, can prove more than 0.833597897.

Our result passes that ceiling.

► **Theorem 1.1.** *Every convex universal cover for the planar sets of diameter one has area at least 0.8344. That is, $\Lambda \geq 0.8344$.*

The proof separates into an analytic part and a single finite computation, and we keep the two apart throughout. Sections 2–4 can be read without a computer. They fix the test sets, prove the ceiling just quoted, confine the placements to a compact box, and establish an estimate that holds simultaneously for every placement in a box, which is what makes a finite search possible at all. Section 5 carries out the search, records it as a certificate, and describes the independent verifier that checks it.

The change that gets past the ceiling is a change of test sets, and it costs nothing in the size of the computation. A body of constant width one has diameter one, so bodies of constant width are admissible test sets. The Reuleaux triangle has constant width one and contains the equilateral triangle of side one, since it is that triangle with three arcs added. The Reuleaux pentagon stands in the same relation to the regular pentagon of diameter one.

Replacing each polygon by the Reuleaux polygon built on it therefore enlarges every copy the cover must contain, while leaving the number of bodies, the number of placement parameters, and the structure of the search exactly as they were. We use the disc, the Reuleaux triangle and the Reuleaux pentagon, all of width one.

1.1 Provenance of the test sets

That choice of test sets is not ours, and the attribution is worth stating precisely. A set of diameter one is contained in a curve of constant width one [11], so it is enough to consider bodies of constant width, and among those the Reuleaux polygons are the easiest to construct. Gibbs [9] states both points, observes that Reuleaux polygons appear empirically to be the most effective choice for a given number of test sets, and runs a simulated annealing search on a family of five of them, reporting 0.83699098. He is careful about what that number is, calling it an upper bound on such a lower bound rather than a bound.

That distinction is not a formality, and it is the reason this paper exists. Write M for the smallest hull area attainable by a chosen family of test sets, over all ways of placing them, so that the observation above reads $\Lambda \geq M$. A search of any kind exhibits a placement, and the area of an exhibited placement is an upper bound on $M(S)$. The quantity that has to be bounded from below is therefore bounded from above, and the direction is wrong by construction rather than by accident. A better optimiser does not repair it. Our own optimiser, run on the same family of five bodies, returns 0.837296, and the placement Gibbs exhibits in 2014 already shows M to be smaller than that, so our value would have been unsound to publish as a bound.

What this paper contributes is the missing direction, together with three things needed to obtain it.

1. An estimate that bounds the hull area from below simultaneously for every placement in a box of placements (Lemma 4.7). Certified bounds of this kind have been obtained for the classical polygonal test sets by Xie [22], who carries the witness points through a parameter domain by interval arithmetic. Lemma 4.7 instead exhibits a fixed body inside every placement the box allows, which is what makes an exhaustive search possible for test sets bounded by circular arcs. Gibbs replaces his arcs by fine polygons; we do not approximate the bodies.
2. The bound of Theorem 1.1, exhaustive over the whole placement space and recorded as a certificate of 486,799,600 nodes. Its verification is logically independent of the search: the verifier reads the certificate alone and re-derives every estimate it checks from the definitions, so the correctness of the search procedure is not part of the argument.
3. An exact ceiling for the classical family (Proposition 3.2), which shows the improvement to be a change of test sets rather than a longer search.

1.2 Related work on computer-assisted proofs

Proofs that reduce to a finite computation over a subdivided parameter space have a long history in discrete geometry. The four colour theorem is the standard reference point, proved by Appel and Haken [1, 2] and reproved with a smaller machine-checked case analysis by Robertson, Sanders, Seymour and Thomas [19]. The resolution of the Kepler conjecture by Hales and coauthors [12] combines combinatorial reduction with interval arithmetic over a subdivision, in a pattern close to the one used here. Szekeres and Peters [21] settle the seventeen-point Erdős-Szekeres problem by an exhaustive computation, and Hass and

Schlaflý [15] prove the double bubble conjecture in a symmetric case by a computer-assisted argument. Nearer to the present problem, Fekete, Gurunathan, Juneja, Keldenich, Kleist and Scheffer [8] determine the critical density for packing squares into a disc with a manual analysis completed by interval arithmetic over subdivided cuboids.

Our computation differs from those in one respect worth stating early. We do not use interval arithmetic. The quantity checked at each leaf is the area of the convex hull of finitely many points, each of which is tested for membership in the body it is supposed to lie in, so only two things can perturb it: a point accepted slightly outside its body, and the rounding of the shoelace sum. Both admit elementary bounds, given in Section 5.4, and their sum is 1.72×10^{-9} .

2 Preliminaries

2.1 Test sets

Everything in this paper rests on one lemma, and the lemma is short. A universal cover has to contain a copy of each of our chosen sets. It is convex, so it also contains the convex hull of those copies taken together. We do not get to choose where the copies sit, so we must take the worst arrangement, and even the worst arrangement gives a bound.

► **Definition 2.1.** *The diameter of a bounded set $T \subset \mathbb{R}^2$ is $\text{diam } T = \sup\{|x - y| : x, y \in T\}$. A convex set $K \subset \mathbb{R}^2$ is a universal cover if for every T with $\text{diam } T \leq 1$ there is a rigid motion g of the plane, orientation reversing if required, with $g(T) \subseteq K$. We write Λ for the infimum of $\text{area}(K)$ over all universal covers K .*

► **Definition 2.2.** *For sets $T_1, \dots, T_k$ of diameter at most one, put*

$$M(T_1, \dots, T_k) = \inf_{g_1, \dots, g_k} \text{area}\left(\text{conv}\left(\bigcup_i g_i(T_i)\right)\right),$$

the infimum running over all k -tuples of rigid motions of the plane. Where the family is clear we write simply M .

► **Lemma 2.3 (Test sets).** *For any sets $T_1, \dots, T_k$ of diameter at most one, $\Lambda \geq M(T_1, \dots, T_k)$.*

Proof. Let K be a universal cover. Each T_i has diameter at most one, so by Definition 2.1 there is a rigid motion g_i with $g_i(T_i) \subseteq K$. Then $\bigcup_i g_i(T_i) \subseteq K$, and K is convex, so $\text{conv}(\bigcup_i g_i(T_i)) \subseteq K$. Taking areas,

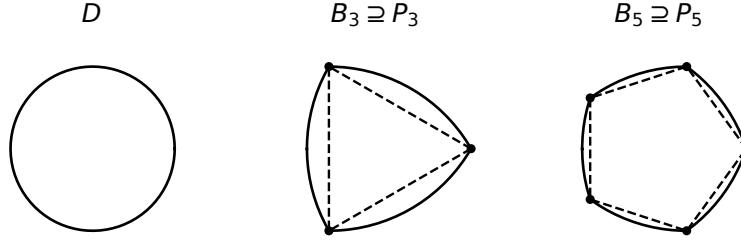
$$\text{area}(K) \geq \text{area}\left(\text{conv}\left(\bigcup_i g_i(T_i)\right)\right) \geq M(T_1, \dots, T_k),$$

the second inequality because the particular motions g_i compete in the infimum of Definition 2.2. The bound holds for every universal cover K , hence for the infimum of their areas. ◀

The two directions are not symmetric. A lower bound on M controls every tuple of motions at once; an upper bound needs only one, since any single tuple competes in the infimum. Section 3 uses the easy direction and Sections 4 and 5 the hard one.

2.2 Constant width

Larger test sets give larger hulls, so we want each T_i to be as large as a diameter-one set can be. Width is the right way to see which sets those are. The width of a convex body



■ **Figure 1** The test sets of Definition 2.8, drawn to a common scale. Left, the disc D of diameter one. Centre and right, the Reuleaux polygons B_3 and B_5 of width one (solid) with the regular polygons P_3 and P_5 of diameter one on the same corners (dashed), the corners marked. The containment is Lemma 2.7.

175 in a given direction is the distance between its two supporting lines perpendicular to that
 176 direction, and the diameter turns out to be the largest width over all directions. A body
 177 whose width is the same in every direction therefore has its diameter equal to that common
 178 width, which is as large as the body can be relative to how far it extends.

179 For a convex body K write $h_K(\theta) = \max_{x \in K} \langle x, u(\theta) \rangle$ for its support function, where
 180 $u(\theta) = (\cos \theta, \sin \theta)$, and $w_K(\theta) = h_K(\theta) + h_K(\theta + \pi)$ for its width in direction θ . K has
 181 constant width w if $w_K \equiv w$.

182 ► **Lemma 2.4.** *A convex body K satisfies $\text{diam } K = \max_{\theta} w_K(\theta)$. In particular a body of*
 183 *constant width w has diameter w .*

184 **Proof.** For $x, y \in K$ with $x \neq y$ set $u = (x - y)/|x - y|$ and let θ be its angle. Then
 185 $|x - y| = \langle x, u \rangle - \langle y, u \rangle \leq h_K(\theta) + h_K(\theta + \pi) = w_K(\theta)$, so $\text{diam } K \leq \max_{\theta} w_K(\theta)$. Conversely
 186 fix θ and choose $x, y \in K$ with $\langle x, u(\theta) \rangle = h_K(\theta)$ and $\langle y, u(\theta + \pi) \rangle = h_K(\theta + \pi)$, which exist
 187 because K is compact. Then $|x - y| \geq \langle x - y, u(\theta) \rangle = w_K(\theta)$, so $\text{diam } K \geq w_K(\theta)$ for every
 188 θ . ◀

189 ► **Remark 2.5.** Every set of diameter at most one is contained in a body of constant width
 190 one [11], so the bodies of constant width one are exactly the maximal test sets and nothing
 191 is lost by restricting to them. Martini, Montejano and Oliveros [17] give a comprehensive
 192 account of such bodies, the completions among them included. We record this because it is
 193 why constant width is the right family to look in. The proof of Theorem 1.1 does not use it.
 194 All that is required there is Lemma 2.4, which certifies that the particular bodies we choose
 195 have diameter one and are therefore admissible in Lemma 2.3.

196 2.3 Reuleaux polygons

197 The Reuleaux polygons are the constant-width bodies that are easiest to write down. Take
 198 an odd number of points, each pair at distance at most one, and intersect the unit discs
 199 centred at them. For the vertices of a regular polygon of diameter one this produces the
 200 familiar rounded shape bounded by circular arcs, and it contains the polygon it was built
 201 from. Figure 1 shows the three bodies we use and the containment.

202 ► **Definition 2.6.** *Let $n \geq 3$ be odd and let $V_1, \dots, V_n$ be the vertices of a regular n -gon*
 203 *of diameter one, so $|V_i - V_j| \leq 1$ for all i, j . The regular Reuleaux n -gon of width one*
 204 *is $B_n = \bigcap_{j=1}^n \overline{D}(V_j, 1)$, where $\overline{D}(c, r)$ is the closed disc of centre c and radius r . We call*

205 $V_1, \dots, V_n$ its corners, and write $R_n = |V_j|$ for their common distance from the centroid, so
 206 that $R_3 = 0.577350269190$ and $R_5 = 0.525731112119$.

207 B_n has constant width one, which is classical [11, 16]. By Lemma 2.4 it has diameter
 208 one, so it is admissible in Lemma 2.3.

209 ► **Lemma 2.7.** B_n contains the regular n -gon $P_n = \text{conv}\{V_1, \dots, V_n\}$ of diameter one on
 210 the same corners.

211 **Proof.** Fix i . For every j we have $|V_i - V_j| \leq 1$, so $V_i \in \overline{D}(V_j, 1)$, and therefore $V_i \in B_n$.
 212 Thus B_n contains every corner, and B_n is convex as an intersection of discs, so it contains
 213 their convex hull P_n . ◀

214 Lemma 2.7 is the whole of the improvement, stated once. Enlarging a test set can only
 215 enlarge the hull of any arrangement containing it, so replacing P_n by B_n increases M , or
 216 leaves it unchanged, for every family in which P_n appears. It does so without changing the
 217 number of test sets or the number of parameters describing their placements.

218 ► **Definition 2.8.** Our test sets are $D = \overline{D}(0, \frac{1}{2})$, the closed disc of diameter one centred at
 219 the origin, together with B_3 and B_5 . The classical family is D together with P_3 and P_5 , the
 220 equilateral triangle of side one and the regular pentagon of diameter one.

2.4 Support functions and hulls

222 Hulls of unions are computed through support functions in Sections 3 and 5, and one identity
 223 carries all of it.

224 ► **Lemma 2.9.** For compact convex sets $K_1, \dots, K_m$, $h_{\text{conv}(\bigcup_i K_i)}(\theta) = \max_i h_{K_i}(\theta)$ for
 225 every θ . If K is translated by t and rotated by ρ about the origin then its support function
 226 becomes $\theta \mapsto h_K(\theta - \rho) + \langle t, u(\theta) \rangle$.

227 **Proof.** A linear functional attains its maximum over $\text{conv}(\bigcup_i K_i)$ at an extreme point of
 228 that hull, and every extreme point lies in some K_i , so the maximum equals $\max_i h_{K_i}(\theta)$.
 229 The second statement is immediate from the definition. ◀

3 The classical family and its ceiling

231 Bounding M from above takes no search. Any single arrangement of the test sets competes
 232 in the infimum of Definition 2.2, so exhibiting one arrangement and measuring its hull settles
 233 the matter. We therefore overestimate the hull deliberately, by intersecting the half-planes
 234 that support it in finitely many directions. That region contains the hull whatever the
 235 directions are, so its area is an upper bound with no approximation argument attached.

236 ► **Lemma 3.1.** Let K be a compact convex set and let $\theta_1, \dots, \theta_N$ be any directions. Then

$$237 \quad K \subseteq Q = \{x \in \mathbb{R}^2 : \langle x, u(\theta_j) \rangle \leq h_K(\theta_j) \text{ for } j = 1, \dots, N\},$$

238 and hence $\text{area}(K) \leq \text{area}(Q)$.

239 **Proof.** If $x \in K$ then $\langle x, u(\theta_j) \rangle \leq h_K(\theta_j)$ for every j , by the definition of the support
 240 function. So $x \in Q$. Area is monotone under inclusion. ◀

241 Taking the θ_j equally spaced makes Q a convex polygon whose vertices are the intersections
 242 of consecutive constraint lines, and its area follows from the shoelace formula. Increasing N
 243 shrinks Q , so the bound improves, and every value obtained along the way remains valid.

244 ► **Proposition 3.2.** $M(D, P_3, P_5) \leq 0.8336$.

245 **Proof.** Place P_3 by the translation $(0.004667984, -0.008121781)$ and P_5 by the rotation
 246 1.464915745 followed by the translation $(0.012090029, -0.020585645)$, leaving D centred at the
 247 origin. By Lemma 2.9 the support function of the hull of the union is the pointwise maximum
 248 of the three support functions, each of which is elementary. Applying Lemma 3.1 with
 249 $N = 1,000,000$ equally spaced directions gives an outer polygon of area at most 0.833597897 ,
 250 the constants here and below being rounded upwards from the computed values. This
 251 arrangement competes in the infimum of Definition 2.2, so $M(D, P_3, P_5) \leq 0.833597897 <$
 252 0.8336 . ◀

253 ► **Remark 3.3.** How the arrangement was found does not enter the proof. We located it by
 254 numerical minimisation, but any arrangement would give a valid bound, and the one quoted
 255 is checked directly. For the same reason the constant is stated with room to spare: the outer
 256 polygon areas at $N = 10^4, 10^5$ and $1,000,000$ are $0.833638744, 0.833601460$ and 0.833597897 ,
 257 a decreasing sequence of valid upper bounds, and rounding to 0.8336 puts the statement far
 258 above any question about the arithmetic.

259 ► **Corollary 3.4.** *Every lower bound on Λ obtained from the test sets D, P_3 and P_5 through*
 260 *Lemma 2.3 is at most 0.8336 , and in particular smaller than 0.8344 .*

261 **Proof.** Lemma 2.3 applied to these three sets yields the single bound $\Lambda \geq M(D, P_3, P_5)$, and
 262 Proposition 3.2 bounds that quantity. ◀

263 These are the test sets of Brass and Sharifi [5] and of Xie [22], and Pál's pair is contained
 264 in them. Corollary 3.4 therefore settles what the classical configuration can deliver. The
 265 published values 0.832 and 0.833 are already within 1.6×10^{-3} and 6×10^{-4} of its ceiling,
 266 and no amount of further computation on those sets reaches 0.8344 . Passing that ceiling
 267 requires different test sets, which is what Sections 4 and 5 do.

268 The same computation applied to our own test sets gives $M(D, B_3, B_5) \leq 0.834781191$,
 269 from the arrangement $(-0.012158832, -0.000000037, 1.256661190, 0.022996607, -0.000003897)$.
 270 That number is not needed for Theorem 1.1. It records how much the family we do use still
 271 has in reserve, and Section 6 returns to the cost of extracting it.

272 4 Reduction to a finite computation

273 Lemma 2.3 turns Theorem 1.1 into a statement about $M(D, B_3, B_5)$, an infimum over an
 274 unbounded family of triples of rigid motions. Three steps make that finite. Symmetry
 275 removes the redundant parameters, an area estimate confines the rest to a compact box, and
 276 an erosion estimate bounds the hull area from below over a whole box of placements at once.
 277 Only the third is specific to this problem.

278 4.1 The gauge

279 The area of a hull does not change when the whole arrangement is moved, so some of the
 280 nine parameters describing three rigid motions carry no information. A global translation
 281 places D at the origin, and D is invariant under rotation about its centre, so D contributes
 282 nothing. A global rotation then fixes the orientation of B_3 . What remains is the translation
 283 of B_3 , and the rotation and translation of B_5 . We call this normalisation *fixing the gauge*,
 284 and it is what reduces nine parameters to five.

► **Definition 4.1.** A placement is a triple $(t_3, \rho, t_5) \in \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^2$, standing for the arrangement of D centred at the origin, B_3 translated by t_3 , and B_5 rotated by ρ about its centre and then translated by t_5 . Here $\mathcal{R}_\rho$ denotes the rotation of the plane through angle ρ about the origin, written this way so that it is not confused with the circumradius R_n of Definition 2.6. Write

$$A(t_3, \rho, t_5) = \text{area}(\text{conv}(D \cup (B_3 + t_3) \cup (\mathcal{R}_\rho B_5 + t_5))).$$

► **Definition 4.2.** A box of placements is a product of five closed intervals, one for each coordinate of Definition 4.1, written through its centre and half-widths as

$$\mathcal{B} = \prod [c - h, c + h] \quad \text{over the coordinates } t_{3x}, t_{3y}, \rho, t_{5x}, t_{5y},$$

with half-widths $h_{3x}, h_{3y}, h_\rho, h_{5x}, h_{5y} \geq 0$. We abbreviate h_x, h_y for the two translation half-widths of whichever body is under discussion.

► **Lemma 4.3.** $M(D, B_3, B_5) = \inf\{A(t_3, \rho, t_5)\}$, the infimum over $t_3, t_5 \in \mathbb{R}^2$ and $\rho \in [0, 2\pi/5)$.

Proof. Given any triple of rigid motions, compose all three with one further rigid motion carrying the image of D to the disc centred at the origin and the image of B_3 to a prescribed orientation. This changes no area, and D is invariant under rotations about its centre, so the resulting arrangement is a placement in the sense of Definition 4.1. Reflections need not be considered separately, since D , B_3 and B_5 each have an axis of symmetry. Rotating B_5 about its own centre by $2\pi/5$ carries it to itself, so ρ may be restricted to $[0, 2\pi/5)$. ◀

4.2 Confining the placements

A test body sitting far from the disc forces a large hull on its own, whatever the rest of the arrangement does, because the hull already contains the disc and a distant point of that body. The area of the convex hull of a disc and a single point is elementary, and it grows with the distance, so beyond some distance the arrangement cannot compete with the bound we are trying to prove and need not be examined.

► **Lemma 4.4.** For $d \geq 1/2$ let

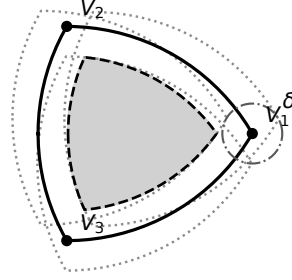
$$f(d) = \frac{1}{4}(\pi - \arccos \frac{1}{2d}) + \frac{1}{2}\sqrt{d^2 - \frac{1}{4}}$$

be the area of the convex hull of D and a point at distance d from its centre. Then f is strictly increasing, and if any point of the arrangement lies at distance $d \geq 1/2$ from the origin then $A(t_3, \rho, t_5) \geq f(d)$.

Proof. The hull of D and a point p at distance d consists of the two tangent segments from p , together with the part of the disc they cut off. Writing $r = 1/2$, the kite bounded by the two tangent segments and the two radii to the points of tangency has area $r\sqrt{d^2 - r^2}$, and the remaining circular sector subtends $2\pi - 2\arccos(r/d)$ and has area $r^2(\pi - \arccos(r/d))$. Adding these and putting $r = 1/2$ gives f , which increases in d because both terms do. The arrangement contains D and the point p , and it is convex, so its area is at least $f(d)$. ◀

► **Corollary 4.5.** If $A(t_3, \rho, t_5) < 0.8344$ then $|t_3| \leq 0.70$ and $|t_5| \leq 0.70$.

Proof. The centre of B_3 lies in B_3 , so after translation it lies at distance $|t_3|$ from the origin. If $|t_3| > 0.70$ then Lemma 4.4 gives $A \geq f(|t_3|) > f(0.70) > 0.836548794 > 0.8344$. The same argument applies to t_5 . ◀



■ **Figure 2** The estimate of Lemma 4.7 for B_3 . Solid, the body $\bigcap_j \overline{D}(V_j, 1)$ with its corners at the centre of a box. Dotted, three of the placements that box allows, each moving every corner by at most δ , the dash-dotted circle showing that budget at V_1 . Shaded, the set $\bigcap_j \overline{D}(V_j, 1 - \delta)$, which lies inside every one of them. Drawn with $\delta = 0.14$ for legibility; the certificate reaches values many orders smaller.

325 The search therefore runs over the compact box $[-0.70, 0.70]^2 \times [0, 2\pi/5] \times [-0.70, 0.70]^2$.

326 ► **Remark 4.6.** Corollary 4.5 is not the sharpest available. Applying Lemma 4.4 to the farthest
 327 corner of a body rather than to its centre uses the fact that the corners already lie at distance
 328 R_n from the centre before any translation. For n corners equally spaced, some corner lies
 329 within angle π/n of any given direction, so $\max_j |\mathcal{R}_\rho V_j + t| \geq (R_n^2 + |t|^2 + 2R_n|t|\cos(\pi/n))^{1/2}$
 330 uniformly in ρ , and the resulting domain is $|t_3| \leq 0.192366$ and $|t_5| \leq 0.195891$, smaller by a
 331 factor of 169 in four-dimensional volume. This recovers the box used by Brass and Sharifi [5].
 332 It does not reduce the cost of the computation, for the reason given in Remark 6.1, and the
 333 certificate on record uses Corollary 4.5.

334 4.3 The erosion estimate

335 The difficulty in searching a box of placements is that each box contains infinitely many of
 336 them. Subdividing helps only if a single computation can settle an entire box, and for that
 337 we need one body that is contained in every placement the box allows. A Reuleaux polygon
 338 is an intersection of unit discs about its corners, and moving the body moves those corners.
 339 If every corner moves by at most δ , then shrinking every disc by δ before moving produces a
 340 body inside all of them at once, because a point within $1 - \delta$ of the original centre is within
 341 1 of the moved one. That is the triangle inequality and nothing more. Figure 2 draws the
 342 situation for B_3 .

343 ► **Lemma 4.7 (Erosion).** *Let $V_1, \dots, V_n$ and $W_1, \dots, W_n$ be points with $|W_j - V_j| \leq \delta$ for*
 344 *every j , where $0 \leq \delta \leq 1$. Then*

$$345 \quad \bigcap_{j=1}^n \overline{D}(V_j, 1 - \delta) \subseteq \bigcap_{j=1}^n \overline{D}(W_j, 1).$$

346 *The left-hand side is non-empty precisely when $1 - \delta$ is at least the circumradius of $\{V_1, \dots, V_n\}$,*
 347 *that is, at least $\min_p \max_j |p - V_j|$.*

Proof. Let p belong to the left-hand side and fix j . Then $|p - W_j| \leq |p - V_j| + |V_j - W_j| \leq (1 - \delta) + \delta = 1$, so $p \in \overline{D}(W_j, 1)$. As j was arbitrary, p lies in the right-hand side. For the second statement, $\bigcap_j \overline{D}(V_j, \varrho)$ is non-empty if and only if some p satisfies $|p - V_j| \leq \varrho$ for all j , which is to say $\min_p \max_j |p - V_j| \leq \varrho$. ◀

We apply this with V_j the corners of a test body positioned at the centre of a box, and W_j its corners at an arbitrary placement in that box. What is needed is a bound on how far a corner can move.

► **Lemma 4.8.** *Let a box of placements in the sense of Definition 4.2 have translation half-widths h_x, h_y and rotation half-width h_ρ . A corner of B_3 , whose orientation is fixed by the gauge, moves by at most $\delta_3 = \sqrt{h_x^2 + h_y^2}$ from its position at the centre of the box. A corner of B_5 moves by at most $\delta_5 = \sqrt{h_x^2 + h_y^2} + 2R_5 \sin(h_\rho/2)$.*

Proof. A translation differing from the box centre by (Δ_x, Δ_y) displaces every point by at most $\sqrt{h_x^2 + h_y^2}$. A rotation by Δ_ρ carries a point at distance R_5 from the body centre through a chord of length $2R_5 \sin(|\Delta_\rho|/2) \leq 2R_5 \sin(h_\rho/2)$. The triangle inequality bounds the composition by the sum. ◀

4.4 The bound on a box

Combining the last three statements gives a single quantity that bounds A from below throughout a box, which is what the search evaluates.

► **Proposition 4.9.** *Let $\mathcal{B}$ be a box of placements as in Definition 4.2, let $V_j^{(3)}$ and $V_j^{(5)}$ be the corners of B_3 and B_5 positioned at its centre, and let δ_3, δ_5 be as in Lemma 4.8. Put $C_n = \bigcap_j \overline{D}(V_j^{(n)}, 1 - \delta_n)$ and*

$$\beta(\mathcal{B}) = \text{area}(\text{conv}(D \cup C_3 \cup C_5)).$$

Then $A(t_3, \rho, t_5) \geq \beta(\mathcal{B})$ for every placement in $\mathcal{B}$.

Proof. Fix a placement in $\mathcal{B}$ and let $W_j^{(n)}$ be the corners of the two bodies as placed. By Lemma 4.8, $|W_j^{(n)} - V_j^{(n)}| \leq \delta_n$ for every j , so Lemma 4.7 gives $C_n \subseteq \bigcap_j \overline{D}(W_j^{(n)}, 1)$, which is the placed body. Hence $D \cup C_3 \cup C_5$ is contained in the union of the three placed bodies, and taking convex hulls and then areas preserves the inclusion. ◀

► **Remark 4.10.** When δ_n is large enough that C_n is empty, which by Lemma 4.7 happens exactly when $1 - \delta_n$ falls below R_n , the proposition still holds with the empty set contributing nothing, and the search falls back on Lemma 4.4. The thresholds are $\delta_3 = 0.422649730810$ and $\delta_5 = 0.474268887881$. Boxes that large occur only at the top of the subdivision.

► **Remark 4.11.** Proposition 4.9 is an inequality about every placement in $\mathcal{B}$, not a sample of them. The subdivision that follows decides only where to stop refining, and no step of the argument treats a finite set of placements as representative of the box that contains it.

5 The certified computation

Proposition 4.9 settles a whole box at once, so the remaining task is to cover the domain of Corollary 4.5 by finitely many boxes on each of which the bound is good enough. That is a subdivision, and it either finishes or it does not. This section describes the subdivision, the record it leaves behind, and the separate program that checks the record.

Throughout, $\tau = 0.8344$ denotes the value being certified and $\mathcal{D} = [-0.70, 0.70]^2 \times [0, 2\pi/5] \times [-0.70, 0.70]^2$ the domain.

5.1 The subdivision

► **Proposition 5.1.** *Suppose $\mathcal{B}_1, \dots, \mathcal{B}_N$ are boxes of placements with $\mathcal{D} \subseteq \bigcup_i \mathcal{B}_i$ and $\beta(\mathcal{B}_i) \geq \tau$ for every i . If in addition $\tau \leq f(0.70)$, then $\Lambda \geq \tau$.*

Proof. Let (t_3, ρ, t_5) be any placement. By Lemma 4.3 we may take $\rho \in [0, 2\pi/5)$, so only the translations can leave $\mathcal{D}$. If the placement lies outside it then $|t_3| > 0.70$ or $|t_5| > 0.70$, and Lemma 4.4 gives $A \geq f(0.70) \geq \tau$. Otherwise it lies in some $\mathcal{B}_i$, and Proposition 4.9 gives $A \geq \beta(\mathcal{B}_i) \geq \tau$. So $A \geq \tau$ everywhere, hence $M(\mathcal{D}, \mathcal{B}_3, \mathcal{B}_5) \geq \tau$ by Lemma 4.3, and Lemma 2.3 gives $\Lambda \geq \tau$. ◀

The covering is produced by bisection. A box whose bound already reaches τ is discarded, since Proposition 5.1 asks nothing more of it. Any other box is split into two, and the two are treated in the same way. Which coordinate to split matters for efficiency and not for correctness: we split the one on which the box is widest, measuring the rotation coordinate by the distance a corner of $\mathcal{B}_5$ travels, so that h_ρ is compared as $R_5 h_\rho$ against the translation half-widths.

► **Definition 5.2.** *The split of a box $\mathcal{B}$ halves its widest coordinate in the weighting just described, producing two boxes whose centres are displaced by $h/2$ and whose half-width in that coordinate is $\frac{h}{2}(1 + \varepsilon)$ with $\varepsilon = 10^{-11}$, the other coordinates unchanged.*

► **Lemma 5.3.** *The two boxes produced by Definition 5.2 cover $\mathcal{B}$.*

Proof. Write the split coordinate as $[c - h, c + h]$. The children carry that coordinate over the intervals centred at $c \pm h/2$ of half-width $\frac{h}{2}(1 + \varepsilon)$, namely $[c - h - \frac{h\varepsilon}{2}, c + \frac{h\varepsilon}{2}]$ and $[c - \frac{h\varepsilon}{2}, c + h + \frac{h\varepsilon}{2}]$. Their union contains $[c - h, c + h]$, and every other coordinate is unchanged. ◀

The widening by ε makes the covering hold with room to spare rather than exactly, so that the conclusion does not rest on two intervals meeting at a single shared endpoint.

► **Corollary 5.4.** *The boxes at which a run stops cover the box it started from.*

Proof. Induction on the subdivision. A box that is discarded covers itself. A box that is split is covered by its two children by Lemma 5.3, and each child is covered by the boxes at which its own subtree stops, by the inductive hypothesis. ◀

Corollary 5.4 supplies the covering hypothesis of Proposition 5.1, so what remains to be checked is only that every box at which the run stopped satisfies $\beta \geq \tau$.

5.2 The certificate

Storing the boxes examined is unnecessary. The split of Definition 5.2 is a deterministic function of a box, so the subdivision is recovered from its shape alone, and that shape is a binary tree.

► **Definition 5.5.** *The search tree of a run has the domain box at its root; a node is a box examined during the run; a node is a leaf if it was discarded, and otherwise has exactly the two children given by Definition 5.2. The certificate is the sequence of bits obtained by*

traversing the tree in depth-first pre-order and emitting 1 at each internal node and 0 at each leaf, together with a header recording τ , the domain half-width, ε , and the resolution parameters of Section 5.3.

A reader given the header and the bits regenerates every box: the root is determined by the header, and each 1 instructs the reader to apply Definition 5.2 and descend. No box is stored, so the record costs one bit per node.

In practice the subdivision is run as 4,194,304 independent subtrees, obtained by splitting the root 22 times before any bound is evaluated, and their bit blocks are concatenated in order. This changes nothing about the tree and allows the work to be divided. It gives a forest rather than a single tree, which supplies a consistency check: a binary tree in which every internal node has two children has one more leaf than it has internal nodes, so a forest of s such trees with L leaves in total has exactly $2L - s$ nodes.

For $\tau = 0.8344$ the emitting run produced 486,799,600 nodes and 245,496,952 leaves in 591.1 minutes on eight cores, satisfying the identity above, and no box reached the refinement guard of 10^{-5} at which the run would have been reported as incomplete. The tree itself is 486,799,600 bits, or 60.85 megabytes. The file adds an index of the 4,194,304 block lengths, a further 16.78 megabytes, and pads each block to a byte boundary, giving 81.3 megabytes on disk and 1.34 bits per node.

A second run of the same search, reported in Table 1, prunes at $\tau + 10^{-9}$ instead and writes no certificate. It closes on 450,922,384 boxes in 364.9 minutes. The two trees differ because the pruning rules differ; the certificate on record is the first.

5.3 Verification

Checking the certificate means recomputing, at every leaf, a quantity that is provably at most $\beta(\mathcal{B})$ and confirming that it still reaches τ . Only elementary operations are used. The disc is replaced by an inscribed polygon, each eroded set by finitely many points of it, and the area of the resulting convex hull is evaluated by the shoelace formula. Every point used is tested against the definition of the set it is supposed to lie in, so the collection is an inner approximation whatever the geometry does.

► **Definition 5.6.** A witness set for a convex set C is a finite subset of C . Given a box, the verifier forms the witness set consisting of the vertices of the regular 1024-gon inscribed in D , together with, for each of C_3 and C_5 , at most 1024 points constructed on the boundary of that set and retained only if they satisfy $|p - V_j^{(n)}| \leq 1 - \delta_n$ for every j .

► **Lemma 5.7.** Let W be a witness set formed as in Definition 5.6 and let $\alpha(W) = \text{area}(\text{conv } W)$. Then $\alpha(W) \leq \beta(\mathcal{B})$.

Proof. The vertices of an inscribed polygon lie in D . A retained point p satisfies $|p - V_j^{(n)}| \leq 1 - \delta_n$ for every j , which is exactly the condition $p \in C_n$ of Proposition 4.9. So $W \subseteq D \cup C_3 \cup C_5$, hence $\text{conv } W \subseteq \text{conv}(D \cup C_3 \cup C_5)$, and areas are monotone. ◀

An inner polygon spanned by points certified to lie in the test sets is the device Xie [22] uses to bound the hull area from below on a parameter domain, with the points carried as functions of the placement by outward-rounded interval arithmetic and their cyclic order certified separately across the domain. Lemma 4.7 reaches the same end by other means. It replaces each moving body by a fixed subset common to every placement in the box, so the witness points do not vary with the parameters and no ordering has to be maintained across

the box. That is what admits test sets bounded by circular arcs rather than by finitely many vertices.

Combining Lemma 5.7 with Proposition 4.9, a leaf at which $\alpha(W) \geq \tau$ satisfies $\beta(\mathcal{B}) \geq \tau$, which is what Proposition 5.1 requires. The retention test is what carries this argument, and it cannot be replaced by an appeal to how the boundary points were constructed: the construction traverses an arc whose angular span exceeds π when $1 - \delta_n$ equals the circumradius of the corners, and points produced there lie outside C_n . Testing each point removes any dependence on that geometry.

► **Remark 5.8.** Two values must be kept apart. The subdivision was run against a threshold slightly above τ , chosen to anticipate the loss in Lemma 5.7 and so to make the later check succeed. That threshold is a heuristic and carries no part of the proof. What carries the proof is the verification, which tests every leaf against τ itself. On this certificate the threshold was in fact chosen too small to guarantee the outcome in advance, and the verifier records as much before proceeding; every one of the 245,496,952 leaves cleared τ regardless, the smallest margin being 1.2409×10^{-5} .

5.4 Arithmetic

The quantity checked at a leaf is the area of the convex hull of finitely many tested points, which limits how floating point can affect it. Two effects are possible and both are bounded outright, so no appeal to the size of the observed margin is needed.

A point is retained when its computed distance satisfies $|p - V_j^{(n)}| \leq 1 - \delta_n + \vartheta$ with $\vartheta = 10^{-12}$, and a distance computed as a square root of a sum of two squares carries relative error at most $4\mathbf{u}$ with $\mathbf{u} = 2^{-53}$. The corners and arc points entering that distance are themselves computed from the sine and cosine of the box-centre rotation, and library trigonometry is not correctly rounded in general, so a further $8\mathbf{u}R$ is allowed for their coordinates, with $R = 0.70 + R_3$. A retained point can therefore lie outside C_n by at most $\varepsilon_0 = 1.0016 \times 10^{-12}$, the trigonometric term sitting three orders below ϑ and not moving ε_0 at this precision. A set within ε_0 of a convex region has hull area exceeding that of the region by at most $\varepsilon_0 P + \pi \varepsilon_0^2$, where P is a bound on the perimeter, and the hull lies within distance $0.70 + R_3$ of the origin so $P \leq 8.026$. This contributes at most 8.04×10^{-12} . The shoelace sum over N vertices contributes at most $\gamma_N S$ with $\gamma_N = N\mathbf{u}/(1 - N\mathbf{u})$ and S the sum of the magnitudes of its terms. Both are bounded structurally rather than sampled. The witness set of Definition 5.6 holds at most $m + 2K = 3072$ points, so $N \leq 3072$; every coordinate satisfies $|x|, |y| \leq R$, so each of the N terms is at most R^2 and $S \leq NR^2 = 5012$. The contribution is at most 1.71×10^{-9} .

Writing $\eta = 1.72 \times 10^{-9}$ for the sum, every verified inequality holds with $\alpha(W)$ replaced by $\alpha(W) - \eta$. The smallest margin attained was 1.2409×10^{-5} , larger than η by a factor of 7.2×10^3 , so Proposition 5.1 applies unchanged. Recomputing the leaf bound in 60 digit arithmetic on the smallest boxes of the certificate reproduces the double precision value to 6.66×10^{-15} , consistent with a bound of η .

5.5 What the verification establishes

The verifier reads the certificate and the header and derives everything else, sharing no code with the subdivision, so Proposition 5.1 is established from the certificate alone.

Several further checks bear on whether the implementation computes what the preceding sections describe. The bound of Lemma 5.7 is implemented a second time against an independent convex hull routine, and 24,704 leaves of this certificate were recomputed that

way, the largest disagreement being 6.05×10^{-13} . The geometry kernel reproduces Pál's $\pi/8 + \sqrt{3}/4$ to 1.1×10^{-16} and the hull drawn in Figure 2 of Brass and Sharifi [5] to 3.2×10^{-6} , neither constant having entered its construction. It passes 28 property tests, and each family passes 29 further checks on Lemma 4.7 and Definition 5.2 at the box sizes the certificate reaches. The forest identity of Definition 5.5 holds exactly, so no subtree is missing from the record.

A last group tests the verifier itself, in both directions. Certificates altered in their tree bits, truncated, or declaring a larger τ , a smaller domain, or a non-positive ε are refused. Valid certificates are accepted, and so are certificates declaring a finer witness resolution, which by Lemma 5.7 yields a tighter bound rather than an invalid one. All 22 of these behave as required. A program that accepted every input would pass the second half of that test and fail the first; one that refused every input would do the reverse.

Proof of Theorem 1.1. Take $\tau = 0.8344$. The boxes at which the subdivision stopped cover $\mathcal{D}$, by Corollary 5.4. At each of them the verification evaluates $\alpha(W)$ and finds $\alpha(W) \geq \tau$, so $\beta \geq \tau$ there by Lemma 5.7, and the arithmetic of Section 5.4 leaves this conclusion intact. Since $f(0.70) > 0.836548794 > \tau$, Proposition 5.1 applies and gives $\Lambda \geq \tau$. ◀

6 Scope of the method

Two questions decide how far this approach reaches. How much of the remaining gap can be taken from the present test sets, and what happens if a fourth body is added. Both are settled by measurement rather than by estimate, and the answers are different in kind: the first is a matter of cost, the second is not.

6.1 Cost against margin

The subdivision must separate every placement from the value being certified, so its cost is governed by the margin μ between τ and the smallest hull area the family attains. Table 1 takes that reference point to be a numerically located optimum rather than the bound of Proposition 3.2, the two differing by under 10^{-8} in the first family, which is far below the range of μ tabulated. As τ approaches the ceiling the boxes near the minimising placement must be refined further before their bound clears τ , and the count grows without bound. Running the same search at a range of targets measures how fast.

Table 1 records the counts and Figure 3 plots them. The eight-dimensional family is the steeper of the two by a wide margin, and it is already the more expensive at every margin the two ranges share.

A power law $N \sim \mu^{-p}$ fits each family, with $p = 2.66$ over the five-dimensional range and $p = 5.50$ over the eight-dimensional one. A rough expectation is $p = d/2$, on the grounds that the bound fails only within a neighbourhood of the minimising placement whose linear size scales as $\sqrt{\mu}$; that gives 2.5 and 4.0 respectively. The five-dimensional fit sits close to it and the eight-dimensional fit exceeds it.

The exponent is not constant. Between successive rows of the five-dimensional column it rises from 3.50 to 3.60 and then falls to 1.94 between the two tightest, so the neighbourhood in question is not self-similar across scales and a single power law understates the cost at wide margins and overstates it at narrow ones. Any extrapolation towards the ceiling using a fit taken over the whole range is therefore conservative, which is the direction that matters when the conclusion is that something is unaffordable.

$D, B_3, B_5 \quad (d = 5)$			$D, B_3, B_5, B_7 \quad (d = 8)$		
τ	μ	boxes	τ	μ	boxes
0.8	0.034781	2,836	0.806495	0.030000	375,104
0.815	0.019781	20,496	0.812495	0.024000	913,876
0.822	0.012781	98,784	0.816495	0.020000	3,379,860
0.826	0.008781	345,964	0.820495	0.016000	11,151,480
0.829	0.005781	1,252,876	0.823495	0.013000	33,943,512
0.831	0.003781	4,075,580	0.825495	0.011000	82,262,500
0.8325	0.002281	14,078,604			
0.8344	0.000381	450,922,384			

■ **Table 1** Boxes examined by the subdivision at a range of targets τ . Every row is the search of Definition 5.2 pruning at $\tau + 10^{-9}$ and writing no certificate, so the last row is the corroborating run of Section 5 rather than the emitting one. The margin μ is measured against a numerically located optimum, 0.834780947 in the first family and 0.836494901 in the second, not against the proved bound of Proposition 3.2; in the first family the two differ by under 10^{-8} . Throughput was 19,648 boxes per second and 11,351 respectively, on eight cores. Generated from `scaling.log`.

6.2 What remains in the present family

Proposition 3.2 applied to D, B_3, B_5 gives a ceiling of 0.83479, so a margin of about 3.8×10^{-4} is unclaimed. Extrapolating from the last row of Table 1 at the local exponent 1.94, itself a two-point estimate from the two tightest margins, certifying 0.8346 would require roughly 1.9×10^9 boxes, near 27 hours at the measured throughput. Nothing in the argument changes; only the subdivision grows. We have not carried this out.

► **Remark 6.1.** The sharper domain of Remark 4.6 does not reduce this cost. Run at $\tau = 0.833$ with every other parameter fixed, the subdivision examines 25,425,568 boxes over the domain of Corollary 4.5 and 27,350,260 over the smaller one. The two a priori estimates agree wherever the hull term in Proposition 4.9 is the larger of the two, which is the case throughout the neighbourhood of the minimising placement, and it is that neighbourhood the box count is made of. A smaller domain removes boxes far from the disc, which the subdivision was already discarding one at a time.

6.3 A fourth test body

Adding the Reuleaux heptagon raises the attainable value to 0.836494901, this one located numerically rather than bounded as in Proposition 3.2, and raises the parameter count from five to eight, which is where the method stops. Taking the eight-dimensional fit of Table 1 at face value, certifying $\tau = 0.8344$ in that family calls for about 8.6×10^{11} boxes, near 880 days at the measured throughput. Allowing for the flattening seen in the five-dimensional column reduces that figure substantially without approaching feasibility.

The obstruction is cost and not principle. Every statement in Sections 2–4 holds verbatim for any finite family of Reuleaux polygons, Lemma 4.7 included, and the subdivision would terminate. What fails is that the number of boxes required grows faster than the dimension is worth. Reaching the value that Gibbs [9] estimates for a family of five bodies would need eleven parameters, further still.

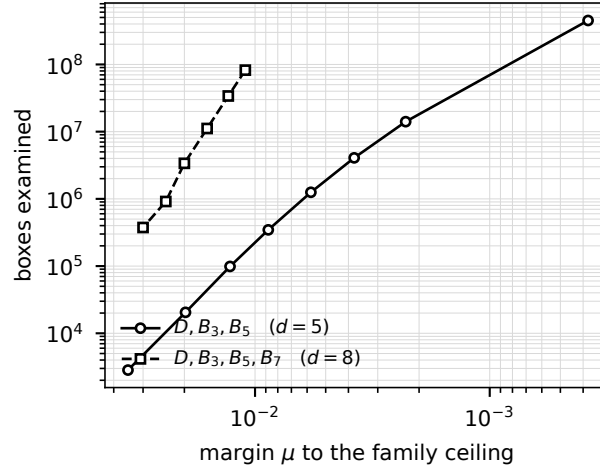


Figure 3 Boxes examined against the margin μ to the family ceiling, for the two families of Table 1, on logarithmic axes with μ decreasing to the right. The certified bound of Theorem 1.1 is the rightmost point of the five-dimensional series.

7 Related problems

The division of labour in the proof is worth stating plainly. Sections 2–4 are analytic and contain no computation: the admissibility of the test sets rests on Lemma 2.4, the ceiling of Proposition 3.2 on one exhibited arrangement, and the reduction on Lemmas 4.4, 4.7, and 4.8. The computation enters once, in Section 5, as a subdivision of one compact box, and what it establishes is the hypothesis of Proposition 5.1 and nothing else.

Two of our constants are sharp for the method and no advance in certification changes them. Proposition 3.2 bounds what the classical test sets can yield at 0.8336, and the same argument bounds our own family at 0.83479. These are properties of the test sets, not of the search, so a faster machine or a better subdivision leaves them where they are. Passing either requires enlarging the family, and the constant-width bodies are already the maximal test sets by Remark 2.5, so the only remaining freedom is to use more of them or to use ones that are not regular.

Against that, the obstruction to going further is cost rather than principle, and the two should not be confused. Within the present family a margin of about 3.8×10^{-4} is unclaimed, and Section 6 prices it. For a fourth test body every statement in Sections 2–4 holds verbatim, the subdivision terminates, and only the box count stands in the way. Brass and Sharifi [5] record the same barrier for their polygonal family and leave it as the obstacle to further progress; Table 1 measures where it now sits.

Lemma 4.7 is not specific to Reuleaux polygons. Its proof uses nothing about the points V_j beyond their number being finite, and in fact not even that: for an arbitrary index set the same triangle inequality gives $\bigcap_i \overline{D}(V_i, 1 - \delta) \subseteq \bigcap_i \overline{D}(W_i, 1)$ whenever $|W_i - V_i| \leq \delta$. A body of constant width one is the intersection of the closed unit discs centred at its own points [11], so the estimate applies to every admissible test set of Remark 2.5, not only to the ones used here. Irregular Reuleaux polygons form an $(n - 3)$ -parameter family at fixed n , and optimising over shape as well as position is a natural next question that the present method can express without modification, at a cost this paper does not attempt.

What would change the picture is a way to bound M from below without exhausting the placement space. Every lower bound for this problem since 1920, ours included, controls all

placements at once by subdividing them, and the cost of that is what Table 1 records. A dual or variational argument, certifying a minimum without enumerating a neighbourhood of it, would remove the dimensional barrier rather than push it, and would reach the families whose numerical values Gibbs [9] reports. The gap $[0.8344, 0.8440935944]$ stands until one or the other side moves.

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